312 m interim results (27 % of the 312 m set)

8 Nov 2026. Prepared by Claude Code (AI assistant) on behalf of Naser Mahfouz. We have 17 of the 72 runs at 312 m: every sed and suppression setting at Nd 20, 50 and 100, with 3D turbulence off. The exception is Nd 100, sed×2, suppression off, which is still running. All runs are 12 h; means are over hours 10–12. Below they are compared with the same settings at 2.5 km, 1.25 km and 625 m. Nd 200–1000 and 3D turbulence on are still running, so the high-Nd branch is not covered yet.

Findings so far

1. **No convergence at 312 m. The step from 625 m to 312 m is the largest yet** (Fig. **??**). With sed×1 and suppression on at Nd 100:

   - LWP drops from 119 to 76 g m^{-2} (163 at 2.5 km).
   - Cloud cover drops from 0.92 to 0.83.
   - w_e rises from 7.0 to 8.1 mm s^{-1}.
   - LWP spatial SD drops, and the cells shrink with Δx (Fig. **??**).

   At Nd 20 the deck breaks up into scattered small clouds: cover 0.49, LWP 39 g m^{-2}.

2. **The low-Nd susceptibility barely changes with resolution.** $S_{20\to50}$ with suppression on is 0.56/0.73/0.97 for sed ×0.5/1/2 at 312 m. That is within ±0.1 of the 2.5 km–625 m values (Fig. **??**).

3. **The LWP peak moves to lower Nd as Δx shrinks.** With sed×1 and suppression on, $S_{50\to100}$ falls from 0.43 at 2.5 km to 0.42, 0.29 and **0.08** at 312 m. So at 312 m LWP is nearly saturated by Nd 100. Whether it falls beyond that waits on the Nd 200–1000 runs.

4. **Precip suppression matters less at 312 m.** At Nd 100, LWP with suppression on and off is 76 vs. 72 g m^{-2}, against 163 vs. 138 at 2.5 km. With suppression off, S is negative on both low-Nd branches at 312 m (−0.11 and −0.13).

5. **Steadiness.** LWP is still thinning slowly over hours 10–12 at 625 m and 312 m alike (−1 to −4 g m^{-2} h^{-1}; Fig. **??**).

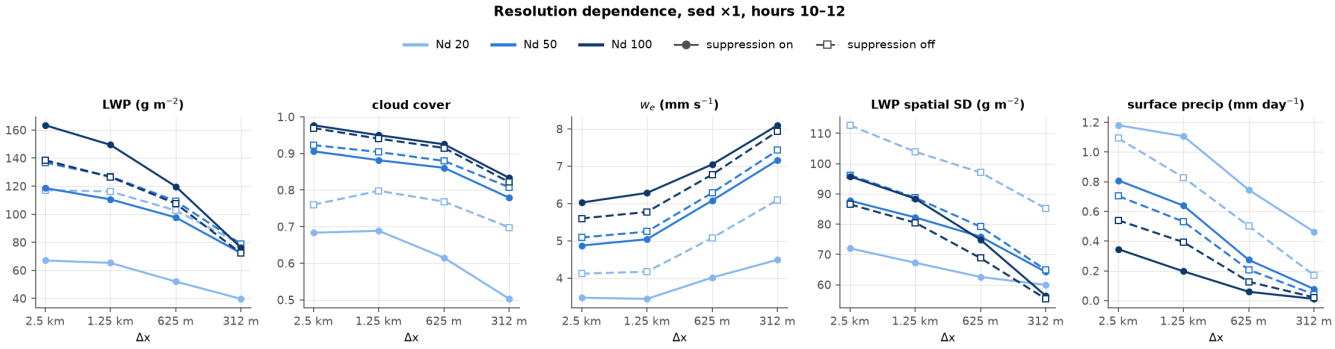


Figure 1: Hour 10–12 means against Δx for sed×1, with Nd 20/50/100 and suppression on (solid) and off (dashed).

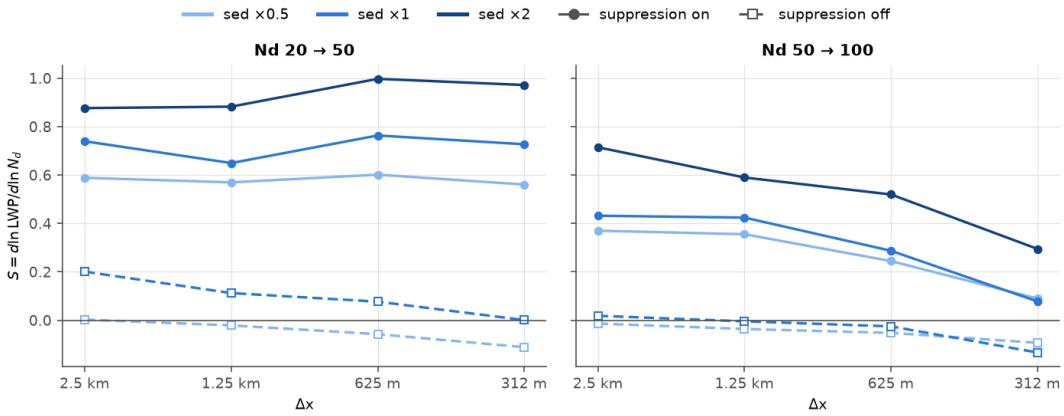


Figure 2: Low-Nd susceptibility against Δx for all sed values.

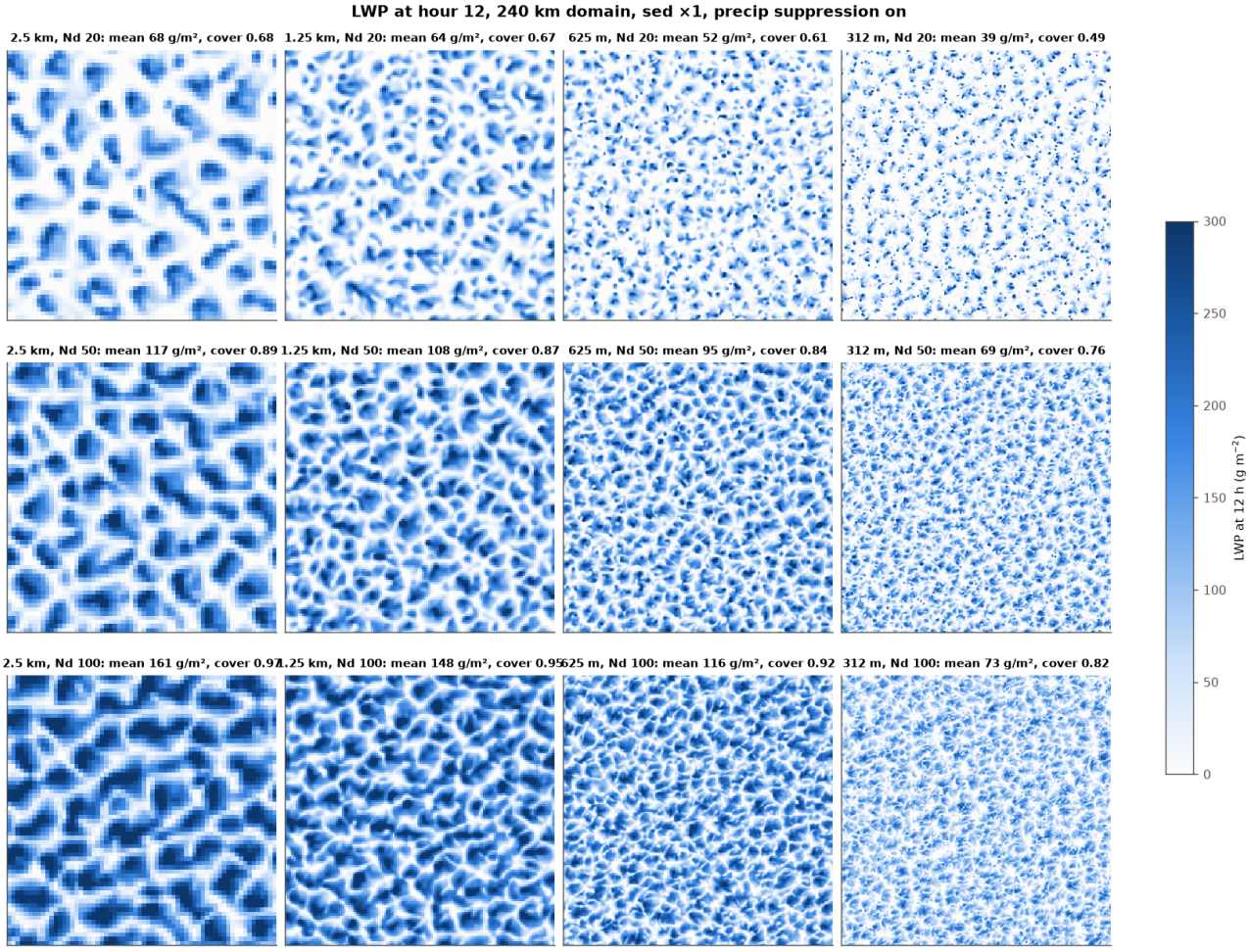


Figure 3: LWP at hour 12 over the full 240 km domain, sed×1, suppression on. Rows: Nd 20/50/100. Columns: Δx.

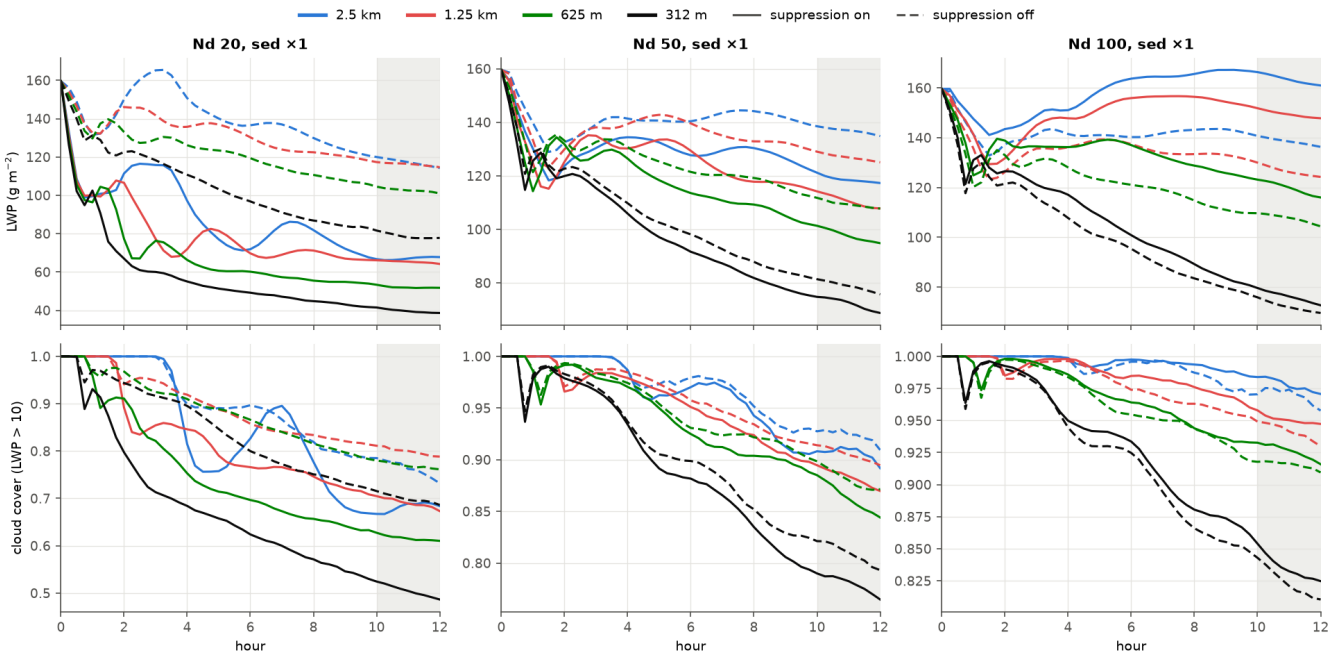


Figure 4: Domain-mean LWP and cloud cover against time at 2.5 km–312 m, sed×1.

Data: `/pscratch/sd/m/mahf708/dp_screamxx_sedps_12h/analysis_312m_interim/` holds `summary_h10-12.csv` for the 17 finished 312 m runs, `susceptibility_lowNd.csv`, the figures (including `figI3_lwp_maps_psoff.png`) and this note. 312 m progress: 233 of 864 model-hours. The first 16-node regular job is expected to start about 01:40 tonight.